

Flying into Sion

A 6° glide path that ends in a visual turn between mountains, an approach that is closed to crews without a specific qualification, a military training area that can reject the flight plan before it is filed, and 2,000 m of runway at the bottom of a valley. What a business jet operator needs to know before committing to LSGS. Written in English.

AERODROME ELEVATION

1,582 ft

THR 25; THR 07 at 1,575 ft

RUNWAY 07/25

2,000 × 40 m

Asphalt · LDA 1,935–1,940 m

IGS RWY 25

6.0° / 4.0°

Instrument, then visual segment

APPROACH CATEGORIES

A / B / C

Crew, operator and aircraft
authorisation required

ORIENTATION

Why Sion is not a normal alpine airport

Plenty of European airports sit in valleys. Sion is unusual because the terrain does not just raise the minima — it changes the shape of the procedure. The instrument approach to RWY 25 is flown on a 6.0° glide path, roughly double the standard, and it does not continue to the runway: it ends at D7.0 ISI, after which the pilot flies visually at 4.0°, offset from the runway axis, turning final over the town.

On top of that geometry, three things decide whether a flight to Sion happens at all: the qualification held by the crew and the operator, the military activity above the valley, and the weather, which here changes faster than a nine-hour TAF suggests.

This briefing is written for operators and crews planning a first visit with a mid-size or large business jet. It summarises how the operation works. It does not replace the

AIP, the NOTAMs, the airport's own qualification material or the aircraft manuals — every figure below must be verified against the current publications.

ACCESS

You cannot simply file and go

IFR approaches and departures at Sion are restricted to pilots, operators and aircraft holding the Sion airport qualification (AIP LSGS AD 2.22). The level held by the pilot in command decides which procedures and which minima are available.

LEVEL	HOW IT IS OBTAINED	WHAT IT ALLOWS, IN BROAD TERMS
Type A	Online exam	Day only, and only with substantially higher decision altitude, visibility and ceiling requirements
Type B	Ground briefing plus a check flight in the aircraft type, with a qualified instructor	Decision altitude driven by aircraft performance, lower visibility, and — with the high-performance departure and RWY 25 in use — night operations

Recency matters: the qualification is tied to the operator and to the aircraft type, and it lapses if the airport is not flown for an extended period. Plan the qualification process weeks ahead of the trip, not days. Simulator credit is accepted for some renewals; the current rules are published by the airport.

PRACTICAL CONSEQUENCE

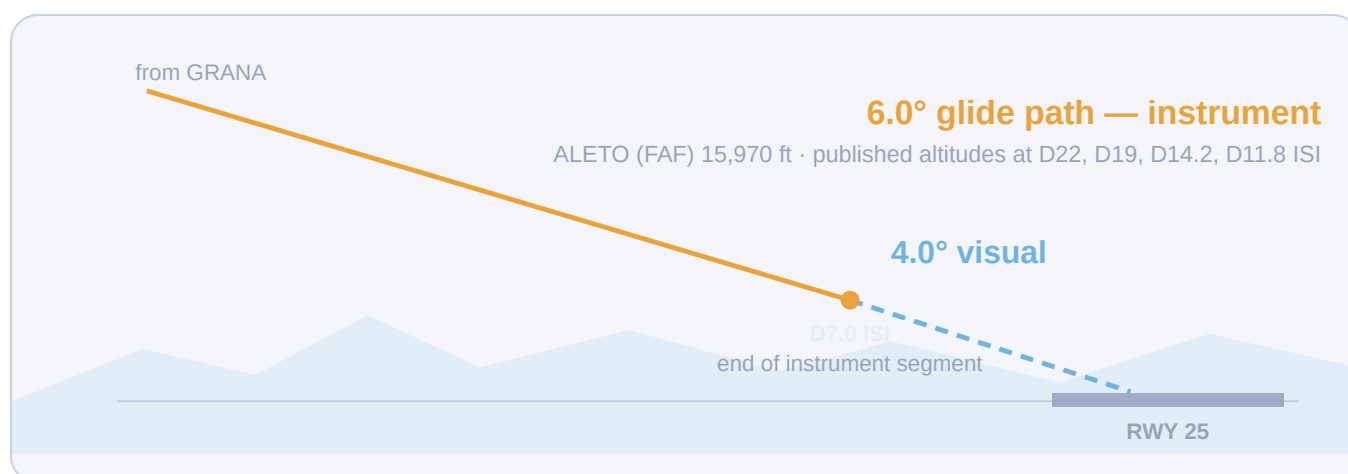
A short-notice charter request for Sion, from an operator without the qualification, is not a scheduling problem — it is a no. The realistic alternatives are Geneva, Bern or Milan, with surface transport into the Valais.

ARRIVAL

The IGS RWY 25 and its visual segment

The approach is an IGS: an ILS-like installation whose localiser is deliberately offset 6.5° from the runway centreline, because the valley does not allow a straight-in

path. The arrival is flown via the RNAV STAR to GRANA, then the procedure descends on the 6.0° glide path to **D7.0 ISI**, about 5.9 NM from the threshold.



Schematic only — not to scale and not for navigation. The published profile, altitudes and decision altitudes are in AIP Switzerland, LSGS AD 2.24.10-1.

From D7.0 ISI the flight continues visually on the offset track, passes abeam St-Léonard about 3.9 NM from the runway, and turns left onto the extended centreline just south of the Sion hospital, following **PAPI 25 at 4.0°** to landing. The PAPI beam is itself aimed 5° north of the runway axis, towards that offset path, so it is not seen from a conventional straight-in position.

- **GNSS is required** for the initial and intermediate segments (RNAV 5), and the approach is prohibited if the glide path is unserviceable.
- The published altitudes on the glide path are **hard constraints**, not advisory steps.
- The runway is frequently **not in sight** at the end of the instrument segment. The visual segment is flown on terrain features and carries no terrain clearance guidance of its own.
- Part of the procedure runs through **Class E airspace**: VFR traffic without radio and without transponder must be expected.

AIRCRAFT CAPABILITY

Most transport-category jets have a maximum autopilot- or flight-director-coupled glide path well below 6°, and an approach-angle limit around 4.5° for the certified landing technique. In practice the steep segment is hand-flown, with a plan for the configuration, the rate of descent (roughly 1,300–1,700 ft/min at typical approach speeds) and the stabilisation gate. Check the figures in your own AFM and operations manual before accepting the approach.

Circling and RWY 07

Circling to RWY 07 exists, but it is limited to **categories A and B**, with a speed limit, special training and no night circling. A category C jet therefore plans on RWY 25 or diverts. Landing on RWY 07 is possible when the conditions allow it, but the runway is short for a heavy jet and there is no separate published missed approach — the low-performance departure profile serves as the visual reference.

DEPARTURE

High performance, low performance,
and what the aircraft can actually do

At Sion, "performance" is not a certification class. It describes whether the aircraft can meet the climb gradient of the steep departure out of the valley, all engines operating.

	HIGH PERFORMANCE (HP)	LOW PERFORMANCE (LP)
Climb requirement	About 13.3% to 10,100 ft, all engines	About 3.7% to 12,500 ft, after a visual segment
Runway	RWY 25 only	RWY 07 and 25 — from RWY 25 restricted to CAT A/B
Day / night	Day and night	Day only
Character	Instrument departure, no visual segment	Initial visual segment in the valley before joining the SID

For a category C jet this usually resolves to the **HP departure from RWY 25**, or the LP departure from RWY 07 — which in turn is the limiting case for take-off weight, because the usable length and the terrain ahead are far less forgiving in that direction.

- Engine-out procedures at Sion are **special, not straight-out** on RWY 25: they follow an escape route along the valley. Run a proper runway analysis for the specific aircraft instead of relying on generic AFM data.
- Maximum take-off weight is often set by the engine-out escape route, not by the runway length.
- On a contaminated runway, count on **no flex thrust and no clearway credit**, and recompute — RWY 07 becomes limiting first.

- Contact ground before start-up; initial climb clearance comes from ATC, and maximum altitudes apply when the military areas are active.
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AIRSPACE

The military area can reject the flight plan

The IFR procedures cross Swiss Air Force training areas. When those areas are active, the restriction is applied at flight-plan level through the non-standard planning zone **LSNPZ1**: IFR flight plans to Sion are rejected for that period, with a message pointing at a forbidden route — not merely delayed by a slot.

- Activity is typically on **weekday mornings and afternoons**, and is published the day before from 16:00 UTC (15:00 UTC in summer) in the Airspace Use Plan. Updates arrive through the UUP.
- ATFM slots, by contrast, appear about two hours before departure — far too late to rebuild a long-haul plan.
- Two ways around a rejection: **move the arrival time** outside the active window, or, in suitable weather, **plan to cancel IFR** at one of the entry fixes and continue VFR below the areas. An aircraft that cannot cancel IFR must divert or hold.

Check the AUP the evening before every flight, and the UUP again before departure. On a long sector, this is the single item most likely to change while the aircraft is airborne.

WEATHER

Valley weather, and the winter picture

- The Sion TAF is valid for **nine hours** only. It will not cover a long inbound sector: plan a self-briefing update en route.
- The **foehn** brings strong, gusty, turbulent southerly flow. With it, the wind aloft and the surface wind can point in different directions, and the approach can become unflyable while the reported surface wind still looks acceptable.
- Summer **thunderstorms** and winter inversions both matter more here than the raw minima suggest, because the visual segment needs the terrain in sight.
- In winter the runway can be contaminated. Take-off and landing performance must be recomputed with the actual runway condition code and contaminant depth.

WINTER PLANNING

Runway de-icing at Sion is carried out by the military base rather than by the airport administration. Expect delays and build them into the slot and crew duty planning. For the aircraft itself, a heated hangar is available and is often the better option: it removes contamination without fluid and keeps the departure independent of the de-icing queue — subject to hangar availability, so it must be requested early.

ON THE GROUND

Handling, customs and parking

Sion has a business-aviation FBO with a private terminal, customs and immigration on site, crew and passenger facilities, and hangarage large enough for heavy business jets. Outside parking and heated hangar space are both available, as are the usual ramp services, lavatory and water servicing, GPU, cleaning and catering coordination.

- Opening hours are limited (HX) and operations outside them need prior permission — in winter, with a deadline the preceding day.
- The rescue and fire fighting category available is limited and can require advance coordination for a large jet. Confirm the category for the planned time.
- Outside parking is charged per night after a short free period; hangar space is charged from the first minute and carries cancellation fees. Current tariffs are published by the handling agent and quoted on request.

BEFORE EVERY FLIGHT

Planning checklist

- ☐ Qualification valid for the pilot in command, the operator and the aircraft type, with recency confirmed.
- ☐ AUP checked the evening before for LSNPZ1; UUP re-checked before departure.

- ☐ NOTAMs read for the approach — procedure altitudes and decision altitudes at Sion have been amended by NOTAM in the past.

- ☐ TAF and TREND checked late, with the nine-hour validity in mind, plus an en-route update plan.

- ☐ Runway analysis and take-off data computed for the actual runway, escape route, temperature and runway condition.

- ☐ Landing distance checked against the available landing distance, with the correct factor and the expected runway state.

- ☐ Alternate selected that is reachable with the fuel on board, with its handling and hours confirmed.

- ☐ Approach briefing covering the steep segment, the offset, the visual turn, the go-around and the terrain escape.

- ☐ Handling, customs, hangar and de-icing arranged, with winter delays built into the schedule.

SOURCES

Where the official information lives

Everything here is a summary. The authoritative documents are free of charge and should be consulted directly for every operation.

AIP Switzerland

skyguide's aeronautical information service: LSGS AD 2, the aerodrome, SID, STAR and approach charts, and the NOTAMs.

Sion airport briefing

The qualification scheme, the training material and the airport's own operational briefing.

EUROCONTROL Network Manager

Airspace Use Plan and updates, route availability, flight-plan validation and ATFM messages.

skyguide AIM services

Special flights, coordination requirements and contact with the Swiss aeronautical information management service.

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